

Cameron Jones — Portfolio Information Sheet

Goal: Build a world-class personal website that grows with my career in AI, robotics, research, and technology.

1. Personal Information

Full Name: Cameron Jones

Professional Title:

(Example: Computer Science Student | AI Researcher | Robotics Engineer)

Current University: Tuskegee University

Major:

Expected Graduation: December 2026

Current Location: Nashville TN

Contact

Email: cameron.kalon@gmail.com

GitHub: [CamJ43737 \(Cameron\)](#)

LinkedIn: [Cameron Jones | LinkedIn](#)

Resume (PDF): (i have my resume PDF i can attach it later i dont know how to attach it to this)

2. Professional Bio

Short Bio (2–3 sentences)

I'm a Computer Science student at Tuskegee University focused on artificial intelligence, robotics, and intelligent systems. My work spans precision agriculture, healthcare technology, and autonomous robotics, where I develop AI-driven solutions that solve meaningful real-world problems through research, software engineering, and innovation.

Long Bio

Tell your story.

Who are you?

Why are you passionate about technology?

I'm passionate about technology as Technology is more than just an interest to me—it is the way I turn ideas into solutions that can make a meaningful difference in people's lives. My passion has grown through experiences developing AI-powered precision agriculture systems, autonomous robots, drones for environmental monitoring, smart transportation technologies, and digital twin applications that support aging in place. What excites me most is solving complex problems by combining artificial intelligence, robotics, computer vision, machine learning, and edge intelligence to address real-world challenges. I enjoy the process of learning new technologies, pushing myself beyond the classroom, and continuously improving my skills, whether that means debugging a difficult system, mastering a new programming language, or designing more innovative solutions. My goal extends beyond building technology for its own sake; I want to create intelligent systems that improve healthcare, strengthen communities, advance scientific research, and enhance everyday life. Ultimately, I aspire to contribute to groundbreaking research and innovation that leaves a lasting impact on society while inspiring others to use technology as a force for positive change. Long before I wrote my first line of code, I was fascinated by creating things.

As a kid, I spent hours building with Legos, constantly imagining how individual pieces could become something much larger. That same curiosity eventually led me to building computers, learning how hardware worked, and exploring the software that powered them.

What started as curiosity evolved into a passion for engineering.

Technology became more than a hobby—it became a way to solve problems.

Today, I'm pursuing a degree in Computer Science at Tuskegee University where I conduct research in artificial intelligence, robotics, healthcare technology, and precision agriculture.

Whether I'm programming autonomous farming robots, designing AI systems for aging-in-place healthcare, developing machine learning tools, or creating intelligent software, my goal remains the same:

Build technology that improves people's lives.

My journey has taken me through research programs at Tuskegee University, Prairie View A&M University, ACCESS-CI at the University of Illinois Urbana-Champaign, and the Minority Serving Cyberinfrastructure Consortium.

Each experience has strengthened my belief that the future belongs to engineers who can bridge software, AI, robotics, and human-centered design.

I'm excited to continue that journey through graduate research and eventually contribute to world-class AI research organizations where I can help build the next generation of intelligent systems.

3. Personal Story

Why did you choose Computer Science?

I chose Computer Science because it allows me to transform ideas into reality. I've always enjoyed building things—from Legos to custom computers—and programming became the ultimate creative tool. Computer Science combines logic, creativity, and engineering into one discipline that enables me to solve meaningful problems.

Why Artificial Intelligence?

Artificial Intelligence fascinated me because it allows machines to learn rather than simply follow instructions. I became interested in how intelligent systems can analyze data, recognize patterns, and make decisions that help people. AI represents one of the most powerful technologies of our generation, and I want to contribute to shaping its future responsibly.

Why Robotics?

Robotics brings software into the physical world. Watching code control real machines—whether autonomous farming robots, drones, or robotic assistants—showed me that software can directly improve people's lives. Robotics combines programming, hardware, sensors, artificial intelligence, and engineering into one discipline, challenging me to think creatively, solve complex problems, and continuously learn. It is this intersection of technology and real-world impact that fuels my passion for robotics and inspires me to develop innovative solutions that make a meaningful difference.

Why Agriculture?

Agriculture is one of the world's oldest industries, yet it faces some of its newest challenges. Through AI Farms, I discovered how artificial intelligence, computer vision, drones, and robotics

can improve food production while conserving resources. That experience showed me that technology can support sustainability on a global scale.

Why Healthcare?

Healthcare presents some of the most meaningful opportunities for artificial intelligence. Through Project AEGIS and the MS-CC Research Internship, I began exploring how AI can improve patient monitoring, support independent living, and assist caregivers through intelligent systems. I want to help create technology that makes healthcare more accessible, proactive, and human-centered.

Why Research?

Research allows me to work on problems that don't already have solutions. I enjoy asking difficult questions, exploring new ideas, and building technologies that have never existed before. Research challenges me to think critically while creating innovations that can benefit society.

Who inspired you?

Honestly my dad inspires me.

My dad inspires me because he has always taught me the importance of discipline, responsibility, and becoming the person you want to be. One of the biggest lessons he taught me is to always take care of what I am given because proper stewardship allows things to last longer. Whether it is my education, my opportunities, my possessions, or my relationships, he showed me that respecting what I have is a reflection of my character. He also taught me that the fastest way to reach a goal is through a straight path—meaning that focus, consistency, and avoiding unnecessary distractions are essential for success.

One of the values my dad passed down to me is the importance of creating my own future. His favorite movie, *The Last Dragon*, represents a message he believes in: being the master of your own fate and taking control of your destiny. That mindset has shaped the way I approach challenges and opportunities. He also encouraged me to play golf because he understood that it is not only a sport but also a place where relationships, networking, and lifelong connections are built. That advice motivated me to join a golf team in high school and taught me the importance of building relationships beyond the classroom.

Most importantly, my dad has always emphasized honesty and integrity. He strongly dislikes dishonesty because he believes trust is one of the most valuable qualities a person can have.

Through his example, he has taught me to be a good steward of my opportunities, stay true to my values, and take responsibility for the path I create. The lessons he has given me continue to influence how I approach my education, my career, and my life.

Biggest challenge you've overcome?

One of the biggest turning points in my life was deciding to leave Fisk University and start a new chapter at Tuskegee University. The decision was not easy because Fisk had become home. I had built friendships, created meaningful connections, and developed a sense of belonging there. Leaving meant walking away from a familiar environment and starting over, which came with a lot of uncertainty.

What ultimately gave me the courage to make the change was my family. My twin brother and I had both attended Fisk, while our twin sisters attended Tuskegee. My sister Caris encouraged me to make the transition, and I still remember how emotional she became when she asked me to come because she truly believed I would love the experience. At the time, I was hesitant, but she was right. Tuskegee challenged me in new ways, introduced me to new opportunities, and allowed me to grow both personally and professionally.

That experience taught me that growth often requires stepping outside of what is comfortable. Leaving Fisk forced me to rebuild—new friendships, new connections, and a new identity—but it also showed me that I am

capable of adapting and creating opportunities wherever I go. Starting over helped me become more confident, resilient, and open to change. It reminded me that sometimes the hardest decisions lead you to the places where you are meant to grow.

Biggest failure?

One of my biggest mistakes was allowing financial considerations to be the primary factor in choosing my college experience. When I was deciding between Fisk University and Tuskegee University, I chose Fisk because I believed I was making the most practical decision financially. At the time, I did not fully understand the many resources, scholarships, programs, and opportunities available to help students fund their education. I focused on the immediate cost rather than the long-term environment, connections, and opportunities that each institution could provide.

I genuinely enjoyed my time at Fisk University and built meaningful relationships there, so I do not view that experience negatively. However, looking back, I realize I may have limited myself by not considering Tuskegee sooner, especially because my twin sisters were already planning to attend and I had an opportunity to be part of that community from the beginning.

That experience taught me the importance of making decisions based on a complete picture rather than only one factor. Financial responsibility is important, but so are mentorship, community, growth opportunities, and finding an environment that aligns with your goals. When I eventually transferred to Tuskegee, I learned that stepping into uncertainty can lead to unexpected growth. It taught me to ask more questions, seek more resources, and never assume that an opportunity is out of reach before exploring every possibility.

Biggest success?

My biggest success is not one specific award, internship,

or accomplishment. I believe my greatest success has been becoming the kind of person and student who continues to create opportunities for himself. My journey has been defined by growth, adaptability, and the willingness to step into unfamiliar spaces.

I went from Fisk University, where I built my foundation and developed meaningful relationships, to Tuskegee University, where I challenged myself to start a new chapter and pursue greater opportunities. Since then, I have continued to build momentum through experiences like AI Farms research, the Prairie View A&M internship, ACCESS-CI, the MS-CC summer research program, research publications, hackathons, and leadership opportunities.

Each experience did not happen by accident. They came from my willingness to learn, take risks, and continuously improve myself. My biggest success has been developing the mindset of someone who does not wait for opportunities to appear but actively seeks ways to grow, contribute, and create impact. The accomplishments are important, but the person I have become through the process is what I am most proud of.

What motivates you every day?

My family

My friends
My dreams and aspirations

Dream Company

OpenAI
Google DeepMind
Microsoft Research
Apple
NASA
AFRL
If i had to choose one it would probably be google

Dream Research Area

Human-centered AI and intelligent robotic systems. It feels more personal

Long-term Career Goal

Lead research and development of next-generation artificial intelligence systems that improve healthcare, agriculture, robotics, and autonomous technologies while mentoring future engineers and advancing responsible AI.

What do you want people to remember you for?

He never stopped building.

4. Education

University

School: Tuskegee University

Degree: Bachelor in Computer Science

Major: Computer Science

Minor (if applicable):

Expected Graduation: 2026

Relevant Coursework: Artificial Intelligence • Machine Learning • Data Structures & Algorithms • Computer Architecture • Computer Vision • Robotics • Embedded Systems & IoT • Software Engineering • Cybersecurity • Database Systems • Linear Algebra • Statistics & Probability

GPA (optional):

Honors: NSF S-STEM Scholar

UNCF Ambassador

TMCF Scholar

Gamma Sigma Delta Honor Society Member

MANRRS SMART Ag Tech Cohort Scholar

HBCUniverse Ambassador

Undergraduate Research Assistant — Tuskegee University AI Farms Initiative

Aspire Leaders Program

Previous Schools

School: Fisk University

Years: 2

Achievements: WEB Dubois Honors Program

5. Resume

Upload your latest resume.

Cant upload here but i can upload later

6. Research Experience

Complete one section for **each** research project.

Research Project #1

Title

AI Farms: Integrating Artificial Intelligence and Robotics for Precision Agriculture

Dates

October 2022 – Present

Institution

Tuskegee University – AI Farms Initiative
CROPPOS (Cornell Robot Precision Planting Systems collaboration)
UIUC (partners)

Advisor
Gregory Bernard
gbernard@tuskegee.edu

Funding Source
USDA / CROPPS / Tuskegee University Research Initiative

Problem

What problem were you trying to solve?

Traditional agriculture faces challenges including inefficient resource usage, labor shortages, and difficulty monitoring crop health at scale. This research explored how artificial intelligence, robotics, drones, and computer vision can improve agricultural decision-making while increasing sustainability.

My Role

Research Assistant / Research Coordinator

- **Program autonomous agricultural robots for planting and watering tasks**
- **Operate NDVI-enabled rover systems and drones for crop monitoring**
- **Collect and analyze agricultural data**
- **Assist with AI-driven environmental monitoring workflows**
- **Present research findings to students, faculty, and external stakeholders**

Technologies Used

Technologies Used

- **Python**
- **Computer Vision**
- **AI / Machine Learning**
- **NDVI Imaging**
- **Robotics Platforms**
- **Drones**
- **Sensors**
- **Arduino**
- **Data Analysis Tools**

Research Methods

Field data collection

Sensor-based environmental monitoring

Autonomous robotics testing

Computer vision analysis

Agricultural data analysis

Results

Improved planting efficiency by approximately 30%

Improved water conservation by approximately 20%

Supported analysis across 15+ acres of farmland

Presented AI agriculture solutions to 25+ stakeholders

Impact

Demonstrated how emerging technologies can support sustainable agriculture by combining AI, robotics, and environmental sensing.

Future Work

Expand autonomous agricultural systems
Improve AI crop prediction models
Integrate additional sensors and robotics platforms

Poster

Integrating AI-Powered Robotic Systems for Precision Agriculture and Environmental Monitoring

Presentation

I will add it i have it tho

Photos

I have the photos they will be labeled one by one and added

Videos

I have a couple of videos added as well as the photos

Repeat this section for:

- Project AEGIS
 - ACCESS-CI
 - CAGI
 - Any additional research
-

Research Project #2

Title

Project AEGIS / AgeXafe XR
Aging-in-Place Digital Twin Ecosystem Using AI-Enabled Monitoring and Robotics

Dates

June 2026 – August 2026

Institution

Minority Serving Cyberinfrastructure Consortium (MS-CC)

Fisk University & Meharry Medical College

Advisor

Landrum, La Chiara M." <llandrum24@mmc.edu>,
"Das, Debashis" <debashis.das@mmc.edu>,
"Chatterjee, Puspita" <puspita.chatterjee@mmc.edu>,
"Ghosh, Uttam" <ughosh@mmc.edu>

Funding Source

NSF Award No. 2401928

Problem

Millions of older adults face challenges maintaining independence while caregivers lack continuous insight into their safety and health. Project AEGIS explores how artificial intelligence, digital twins, and robotics can support safer aging-in-place environments.

My Role

Apartment Framework Lead / AI Research Intern

- **Developed virtual apartment environment in Unity**
 - **Created realistic living spaces for simulation**
 - **Integrated RoboDog autonomous assistance concepts**
 - **Designed navigation pathways and environmental zones**
 - **Supported healthcare AI research workflows**
-

Technologies Used

Technologies Used

- **Unity 6**
- **C#**
- **Python**
- **Artificial Intelligence**
- **Digital Twins**
- **Robotics**
- **NavMesh**

- **3D Modeling**
 - **Data Analytics**
 -
-

Research Methods

- Digital environment simulation
 - Human-centered AI design
 - Robotics testing
 - Healthcare dataset exploration
 - Spatial risk mapping
-

Results

Built apartment digital twin framework
Integrated RoboDog navigation system
Created environment for AI-assisted aging research
Supported MVP development

Impact

Explores how AI and robotics can improve independence, safety, and healthcare support for aging populations.

Future Work

Add AI fall detection
Integrate healthcare sensors

Improve personalized patient models

Poster

I'm adding it later in my pictures folder

Presentation

I will add it i have it tho

Photos

I have the photos they will be labeled one by one and added

Videos

I have a couple of videos added as well as the photos

Research Project #3

Title

ACCESS-CI

Title

AI-Powered Knowledge Retrieval and NLP Automation for Cyberinfrastructure Support

Dates

June 2025 – August 2025

Institution

ACCESS-CI STEP NSF Internship
University of Illinois Urbana-Champaign

Advisor

Dinuka De Silva (Jira) <jira@access-ci.atlassian.net>

Funding Source

USDA / CROPPS / Tuskegee University Research Initiative

Problem

Scientific computing communities require accessible documentation and support systems. The project explored improving chatbot knowledge quality through AI-assisted content processing.

My Role

Software Engineering & NLP Intern

Technologies Used

Python

- **REST APIs**
- **NLP**
- **Markdown**
- **Large Language Models**
- **Jira**
- **Git**

Research Methods

- Coding with python in research environments

Results

- Integrated 50+ verified resources
- Converted 300+ API responses into AI-readable text
- Reduced manual documentation work by 70%
- Improved chatbot input quality by approximately 65%

Impact

Helped improve accessibility of computational resources for researchers and students.

Projects to include:

- AI Farms
 - Project AEGIS
 - ACCESS-CI
 - Prairie View Internship
 - Coca-Cola Internship
 - Personal Website
 - Robot Dog
 - Smart Agriculture
 - Any hackathon projects
 - Personal coding projects
-

8. Publications

Publication Title

9. Awards & Honors

NSF S-STEM Scholar

Year:

2023-Present

Organization:

National Science Foundation

Why received:

Selected based on academic achievement, financial need, and commitment to computer science.

TMCF Scholar

Year:

2024-2025

Organization:
Thurgood Marshall College Fund

MANRRS S².M.A.R.T Academy Cohort

Year:
2026

Organization:
MANRRS

Auburn University Hacks Winner

Year:
2023

Achievement:

Hackathon competition winner.

UIUC

Winner of Precision & Digital Agriculture Hackathon Technology Track.

10. Leadership

UNCF Ambassador

Role:
Student Ambassador

Responsibilities:

- Represent UNCF initiatives
- Promote educational opportunities

- Support HBCU community
-

HBCUniverse Ambassador

Responsibilities:

- Promote HBCU innovation
 - Student engagement
-

Global Fellows Organizer

Responsibilities:

- Leadership development
 - Campus programming
-

NSBE Member

ACM Member

Mayor's Youth Council Ambassador

Year:

2020

Impact:

Developed leadership and civic engagement skills.

Examples:

- UNCF Ambassador
- HBCUniverse

- NSBE
 - ACM
 - MANRRS
 - Mayor's Youth Council
 - Leadership Institute
 - Global Fellows
-

11. Volunteer Experience

- STEM Outreach
 - Tutoring
 - Community Service
 - Church (photography)
 - Coding Camps (gave workshops to teachers on campus and i have pictures)
 - Mentoring
-

13. Skills

Programming

Python
Java
C++
JavaScript
Swift
SQL
HTML/CSS

AI / Machine Learning

TensorFlow
PyTorch
Computer Vision

NLP
LLMs
Data Analysis

Robotics

ROS
Arduino
Raspberry Pi
Sensors
Autonomous Systems
Drones

Cloud

Docker
AWS AI Practitioner

Databases

SQL
[Need others]

Web

HTML
CSS
JavaScript
React [confirm]

Mobile

Swift
iOS Development

Design

Adobe Photoshop
Adobe Premiere Pro

16. Videos

Research demos

Robot videos

Drone footage

Presentations

Campus footage

Interviews

Other

17. Media

Professional headshots

Research photos

Conference photos

Lab photos

Drone photos

Robot photos

Presentation photos

Awards

Internships

Graduation

Campus life

18. Testimonials

Name

Position MANNRS SMART AG Tech track

19. Timeline

Year

Major milestone

Repeat as needed.

Example:

- 2019 – First programming experience
 - 2020 – Mayor's Youth Council
2020 - first began fisk university but later switched to tuskegee because my twin sisters and parents had attended
 - 2022 – Began Tuskegee University
 - 2022 – AI Farms research
 - 2023 – NSF S-STEM Scholar
 - 2024 – Prairie View internship
 - 2026 – Project AEGIS & MS-CC Internship
 - Future – Graduate school, Ph.D., AI research
-

20. Future Goals

Current Projec

Project AEGIS t

Building Now

AI research portfolio website

Robotics systems

Machine learning application

Research Interests

Human-centered AI

Robotics

Computer Vision

Healthcare AI

Precision Agriculture

Autonomous Systems

Dream Labs

OpenAI

Google DeepMind

Microsoft Research

Apple AI Research

NASA

AFRL

Future Publications

Future Technologies You Want to Learn

Humanoid robotics

Advanced AI agents

Autonomous systems

Brain-computer interfaces

Edge AI

21. Website Style Preferences

Websites You Love

1. Apple
 2. Tesla
 3. Linear
 4. Stripe
 5. NVIDIA
 6. OpenAI
-

Favorite Companies

- Apple
 - OpenAI
 - Google
 - NVIDIA
 - Microsoft
 - Tesla
 - NASA
-

Favorite Brands

Brooks

Apple

Favorite Movies

Indiana jones and the temple of doom

The gray man

Bullet train

The matrix

Interstellar

Mission impossible

Favorite Games

Destiny 2

The walking dead

Lego Star Wars the complete saga (Wii)

Lego Indiana Jones the complete saga (Wii)

Pokémon rumble blast (3ds)

Mortal kombat the komplette edition (ps3)

Michael Jackson the experience (Wii)

Wii sports (Wii)

Words you want people to feel when visiting your site:

Innovation

Curiosity

Ambition

Discovery

Future

Impact

Engineering

Purpose

Timeline

I'd expand it considerably.

Year	Milestone
2018	Began building PCs and exploring technology
2019	Learned programming fundamentals
2020	Started at Fisk University; joined the Mayor's Youth Council
2022	Transferred to Tuskegee University to continue a family legacy alongside twin sisters
2022	Joined AI Farms Research Initiative
2023	Became an NSF S-STEM Scholar and expanded leadership involvement
2024	Prairie View A&M research internship; Google CSSI; CAGI Python Bootcamp publication
2025	ACCESS-CI NLP & Software Engineering Internship at the University of Illinois Urbana-Champaign
2026	MS-CC Internship; Project AEGIS; Precision & Digital Agriculture Hackathon Winner

Future Graduate school, Ph.D., AI research, intelligent robotics, and human-centered AI

I also want this implementation:

Let's Connect

Let's
Talk.

Whether you're a researcher, recruiter, or someone building something meaningful at the edge of AI and agriculture — I want to hear from you.

Email MeDownload ResumeLinkedInGitHubPhotography

Send a Message

Fill out the form and I'll get back to you. Whether it's research, collaboration, or just a conversation — I'm listening.

Your Name

Your Email

Reason for Reaching Out

Select one...	Research Collaboration	Job / Internship
Opportunity	Speaking Engagement	Media / Press
Inquiry		General

Message

Send Message

Direct Links

Email

Cameron.kalon@gmail.com

LinkedIn

linkedin.com/in/cameron-jones-5855311b9

GitHub

github.com/CamJ43737

Photography Portfolio

View Photography Work

Resume

Download PDF

Based At

Tuskegee University · Tuskegee, Alabama

During my undergraduate studies at Tuskegee University, I have served as an AI Research Assistant in the AI Farms Research Lab, where I have worked on autonomous robotic systems, precision agriculture, computer vision, drones, LiDAR, and environmental monitoring. I also completed an NSF MS-CC Summer Research Internship at Fisk University and Meharry Medical College, where I contributed to Project AEGIS, an AI-enabled aging-in-place digital twin platform integrating robotics, multimodal sensing, and healthcare technologies. Additionally, I have participated in the Coca-Cola Scholars Internship, Prairie View A&M University's robotics and IoT research program, and the CAGI AI Bootcamp, gaining experience in machine learning, robotics, software development, and interdisciplinary research. These experiences have strengthened my ability to work in collaborative research environments while applying AI to solve real-world problems.

The AFRL Scholars Program would provide an opportunity to apply my background in artificial intelligence, robotics, and data science to national research challenges while learning from leading scientists and engineers. My long-term goal is to pursue graduate studies and conduct research in trustworthy AI, autonomous systems, and intelligent cyber-physical systems for healthcare and defense applications. Working alongside AFRL researchers would allow me to deepen my technical expertise, gain experience with advanced research methodologies, and contribute to technologies with real-world impact.

want to work with AFRL because it represents one of the nation's premier research organizations focused on developing innovative technologies that strengthen national security and improve society. I am particularly interested in AFRL's work involving artificial intelligence, autonomy, computer vision, cybersecurity, and human-machine teaming. The opportunity to contribute to cutting-edge research while collaborating with experienced mentors aligns directly with my career goals of becoming an AI researcher and developing technologies that improve both defense capabilities and quality of life.

Data Structures
Computer Architecture
Discrete Mathematics
Calculus I & II
Linear Algebra
Probability and Statistics
Programming I (Python)
Programming II (Java)
Object-Oriented Programming
Database Systems
Software Engineering
Operating Systems
Artificial Intelligence
Machine Learning
Computer Networks

Python
Java
C++
Swift
HTML/CSS
JavaScript
SQL
Git/GitHub
Linux/UNIX
Windows
macOS
Microsoft Excel
MATLAB
Jupyter Notebook
Visual Studio Code
Xcode

Unity 3D
Arduino
ROS (if applicable)
Computer Vision
Machine Learning
Data Analysis

Artificial Intelligence and Machine Learning
Autonomous Robotics
Computer Vision
Data Analysis and Visualization
Research Design
Scientific Writing
Technical Presentations
Human-Robot Interaction
Sensor Integration
Edge Computing
Problem Solving
Team Collaboration
Software Development
Rapid Prototyping
Literature Review and Technical Documentation

NSF S-STEM Scholarship Recipient
TMCF Scholar
UNCF Ambassador, Tuskegee University
HBCUniverse Ambassador
Gamma Sigma Delta Honor Society
Coca-Cola Scholar Internship Participant
MANRRS SMART Ag Tech Program Participant

Jones, C., et al. Project AEGIS: An AI-Enabled Aging-in-Place Digital Twin for Healthcare Monitoring Using Autonomous Robotic Systems. (IEEE conference paper/manuscript in preparation, 2026.)

Jones, C., et al. CAGI Bootcamp Research: Applications of Artificial Intelligence for Precision Agriculture and Intelligent Systems. (Research paper/manuscript in preparation, 2026.)c